

# MA388 Sabermetrics: Lesson 10

## Value of Plays - Base Hits and Stealing

LTC Jim Pleuss

### Lesson Objectives

1. Compute RE24 to aggregate a player's run value across all plate appearances.
2. Compare batting average and slugging percentage to run-value-based measures of performance.
3. Evaluate the expected value of a stolen base attempt using the run expectancy matrix.
4. Calculate mean run values for singles, doubles, triples, and home runs.

### Review

Last class, we discussed the Run Expectancy Matrix and how to obtain it from the Retrosheet play-by-play data.

```
library(tidyverse)
library(knitr)
library(Lahman)

erm_2011_wide <- read_csv(file = "erm_2011_wide.csv")

kable(erm_2011_wide)
```

| bases | Outs=0    | Outs=1    | Outs=2    |
|-------|-----------|-----------|-----------|
| 000   | 0.4711649 | 0.2546956 | 0.0971845 |
| 001   | 1.4543568 | 0.9374359 | 0.3173913 |
| 010   | 1.0582804 | 0.6501976 | 0.3091673 |
| 011   | 1.9304897 | 1.3388641 | 0.5407407 |
| 100   | 0.8350992 | 0.4960492 | 0.2179536 |
| 101   | 1.7526998 | 1.1496169 | 0.4882597 |
| 110   | 1.4144549 | 0.8739176 | 0.4222569 |

| bases | Outs=0    | Outs=1    | Outs=2    |
|-------|-----------|-----------|-----------|
| 111   | 2.1718266 | 1.4745146 | 0.7610094 |

*Interpret the value in the first row, third column.*

*True or False: The probability of scoring at least one run in the remainder of the inning is 0.25 when there is no one on base with one out.*

*Let's say there is a runner on second base (and no other baserunners). Stealing third base is the most valuable when there are how many outs?*

*Let's say there is a runner on second base with one out. Calculate the expected value of a stolen base attempt if the probability of successfully stealing third base is 70%. In addition, calculate the minimum probability of successfully stealing third base required to make it a better strategy in terms of expected runs.*

## Measuring the Success of a Batting Play

We can estimate the value of a plate appearance as the difference in run expectancy between the new and old states plus the number of runs scored on the play.

$$\text{RUN VALUE} = E[\text{RUNS}]_{\text{new state}} - E[\text{RUNS}]_{\text{old state}} + \text{RUNS}_{\text{scored on play}}$$

Calculate the RUN VALUE of the following plays:

- (1) There are runners on first and second with no outs. The batter successfully sacrifice bunts, resulting in runners on second and third with one out.
- (2) There is a runner on second base with no outs. The batter hits a single that scores the runner from second. The batter stops at first base.
- (3) What is the most valuable play in baseball? Explain.

## Measuring Batting Success Over a Season: David Ortiz

We'll start with all batting events in the 2011 season.

```
# Load the data.
site = "https://raw.githubusercontent.com/maxtoki/baseball_R/"
fields <- read_csv(file = paste(site, "master/data/fields.csv", sep = ""))
retro <- read_csv(file = paste(site, "master/data/all2011.csv", sep = ""),
                  col_names = pull(fields, Header),
                  na = character())
colnames(retro) <- tolower(colnames(retro))

# Add states and run values.
source("./RunExpectancyMatrixFunctions.R")
get_run_expectancy_retro(retro)
retro2011 <- retro_event_run_value
```

Now let's look specifically at David Ortiz.

```
# Get Ortiz's playerID.
ortizID <- People |>
  filter(nameFirst == "David", nameLast == "Ortiz") |>
  pull(retroID)

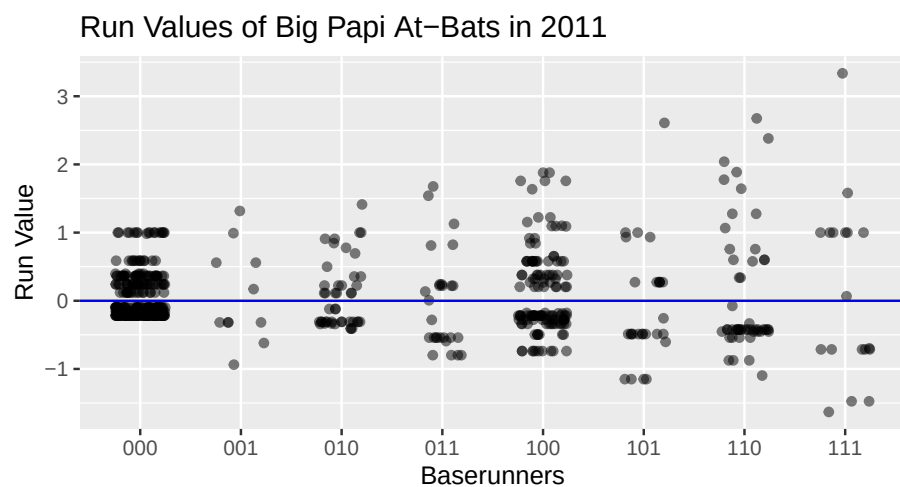
ortiz_df <- retro2011 |>
  filter(bat_id == ortizID,
         bat_event_fl == TRUE)

ortiz_df |>
  select(game_id, inn_ct, state, new_state, run_value) |>
  head(10) |>
  kable()
```

| game_id      | inn_ct | state | new_state | run_value  |
|--------------|--------|-------|-----------|------------|
| ANA201104210 | 2      | 000 0 | 010 0     | 0.5871155  |
| ANA201104210 | 4      | 000 0 | 000 1     | -0.2164694 |
| ANA201104210 | 6      | 000 0 | 100 0     | 0.3639342  |
| ANA201104210 | 7      | 110 2 | 110 3     | -0.4222569 |
| ANA201104210 | 9      | 000 2 | 100 2     | 0.1207691  |
| ANA201104220 | 2      | 000 0 | 000 1     | -0.2164694 |
| ANA201104220 | 4      | 000 1 | 000 2     | -0.1575110 |
| ANA201104220 | 5      | 101 2 | 101 3     | -0.4882597 |
| ANA201104220 | 7      | 010 2 | 010 3     | -0.3091673 |
| ANA201104230 | 2      | 000 0 | 000 1     | -0.2164694 |

Here's a plot of Ortiz's run values by runner state.

```
ortiz_df |>
  ggplot(aes(x = bases, y = run_value)) +
  geom_jitter(width = 0.25, alpha = 0.5) +
  geom_hline(yintercept = 0, color = "blue") +
  labs(x = "Baserunners",
       y = "Run Value",
       title = "Run Values of Big Papi At-Bats in 2011")
```



Next, let's look at total runs in different baserunner situations.

```
runs_ortiz <- ortiz_df |>
  group_by(bases) |>
  summarize(runs = sum(run_value) |> round(1),
            PA = n())

runs_ortiz |> kable()
```

| bases | runs | PA  |
|-------|------|-----|
| 000   | 13.1 | 301 |
| 001   | 0.8  | 11  |
| 010   | 3.4  | 44  |
| 011   | -0.9 | 27  |
| 100   | 14.9 | 127 |
| 101   | -2.2 | 27  |
| 110   | 4.4  | 50  |
| 111   | 2.1  | 18  |

Lastly, we can calculate a statistic called  $RE24$  which sums runs over the different baserunner combinations.

```
runs_ortiz |>
  summarize(RE24 = sum(runs))
```

```
# A tibble: 1 x 1
  RE24
```

```
<dbl>
1 35.6
```

How do we interpret RE24?

Come up with a modification for RE24 that could be used to rank batters according to what they produced given the opportunity they had. How would you do it?

## Part II - Measuring the Success of a Stolen Base Attempt

Your book dedicates roughly three pages to evaluating base stealing. We're going to discuss it briefly and return to our evaluation of batting performance. Your task: Develop an RE24-inspired measure to rank MLB's top base stealers.

Let's start with the same approach to valuing an individual event.

$$\text{RUN VALUE} = \text{RUNS}_{\text{new state}} - \text{RUNS}_{\text{old state}} + \text{RUNS}_{\text{scored on play}}$$

where  $\text{RUNS}_{\text{new state}}$  and  $\text{RUNS}_{\text{old state}}$  are their respective entries in the run expectancy matrix and  $\text{RUNS}_{\text{scored on play}}$  is the number of runs that scored on the play.

```
kable(erm_2011_wide)
```

| bases | Outs=0    | Outs=1    | Outs=2    |
|-------|-----------|-----------|-----------|
| 000   | 0.4711649 | 0.2546956 | 0.0971845 |
| 001   | 1.4543568 | 0.9374359 | 0.3173913 |
| 010   | 1.0582804 | 0.6501976 | 0.3091673 |
| 011   | 1.9304897 | 1.3388641 | 0.5407407 |

| bases | Outs=0    | Outs=1    | Outs=2    |
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| 100   | 0.8350992 | 0.4960492 | 0.2179536 |
| 101   | 1.7526998 | 1.1496169 | 0.4882597 |
| 110   | 1.4144549 | 0.8739176 | 0.4222569 |
| 111   | 2.1718266 | 1.4745146 | 0.7610094 |

Let's say there is a runner on first base with no outs and you are considering bunting. Historically, a bunt in this situation would result in a runner on second with one out 75% of the time, a runner on first and second with no outs 10% of the time, a runner on first with one out 10% of the time, and runners on second and third with no outs 5% of the time. If you have an average hitter at bat, does bunting or hitting away have the highest expected run value?

For more on bunting, see our [textbook author's blog](#).

## RE24

Earlier, we investigated the run value of David Ortiz's 2011 plate appearances. We can aggregate these run values for each player to a single statistic called "RE24".

*If we sum all players RE24 together, what should be the result?*

In baseball, there are [context-dependent](#) and [context-independent](#) statistics. Name some examples of each. Which type of statistic is RE24?

Calculate the RE24, plate appearances, and total starting run expectancy for each player in 2011.

```

re24_2011 <- retro2011 |>
  group_by(bat_id) |>
  summarize(re24 = sum(run_value),
            PA = length(run_value),
            runs_start = sum(rv_start)) |>
  arrange(-re24)

# Add names from the People data frame (note it's not playerID, but retroID).
re24_2011 <- re24_2011 |>
  left_join(select(People, retroID, nameLast, nameFirst),
            by = c("bat_id" = "retroID")) |>
  mutate(name = paste(nameFirst, nameLast, sep = " ")) |>
  select(-nameLast, -nameFirst)

re24_2011 |>
  head(15) |>
  kable(digits = 1)

```

| bat_id   | re24 | PA  | runs_start | name              |
|----------|------|-----|------------|-------------------|
| cabrm001 | 73.9 | 701 | 332.9      | Miguel Cabrera    |
| bautj002 | 73.6 | 693 | 332.8      | Jose Bautista     |
| fielp001 | 65.6 | 734 | 363.6      | Prince Fielder    |
| vottj001 | 64.5 | 743 | 343.3      | Joey Votto        |
| braur002 | 61.9 | 663 | 316.3      | Ryan Braun        |
| kempm001 | 60.1 | 704 | 349.1      | Matt Kemp         |
| ellsj001 | 58.1 | 748 | 347.4      | Jacoby Ellsbury   |
| gonza003 | 57.3 | 755 | 367.3      | Adrian Gonzalez   |
| berkl001 | 55.4 | 600 | 289.4      | Lance Berkman     |
| martv001 | 49.3 | 606 | 318.2      | Victor Martinez   |
| canor001 | 49.1 | 695 | 335.9      | Robinson Cano     |
| napom001 | 46.6 | 444 | 215.0      | Mike Napoli       |
| granc001 | 44.1 | 727 | 362.9      | Curtis Granderson |
| teixm001 | 41.2 | 718 | 347.9      | Mark Teixeira     |
| gorda001 | 41.1 | 727 | 344.3      | Alex Gordon       |

First, let's see if the average is about zero.

```
re24_2011 |>
  summarize(mean_RE24 = mean(re24))
```

```
# A tibble: 1 x 1
  mean_RE24
    <dbl>
1      0.411
```

Now, let's plot RE24 for each player versus starting runs.

```
library(ggrepel)

# Create data frame of points to label.
labels <- re24_2011 |>
  filter(PA >= 502) |>
  filter(re24 > 50 | re24 < -30)

re24_2011 |>
  filter(PA >= 502) |>
  ggplot(aes(x = runs_start, y = re24)) +
    geom_point() +
    geom_smooth(se = FALSE) +
    geom_text_repel(data = labels, aes(x = runs_start,
                                         y = re24,
                                         label = name)) +
    labs(title = "Player RE24 vs. Opportunity",
         x = "Cumulative Starting Run Value for All At-Bats",
         y = "RE24")
```

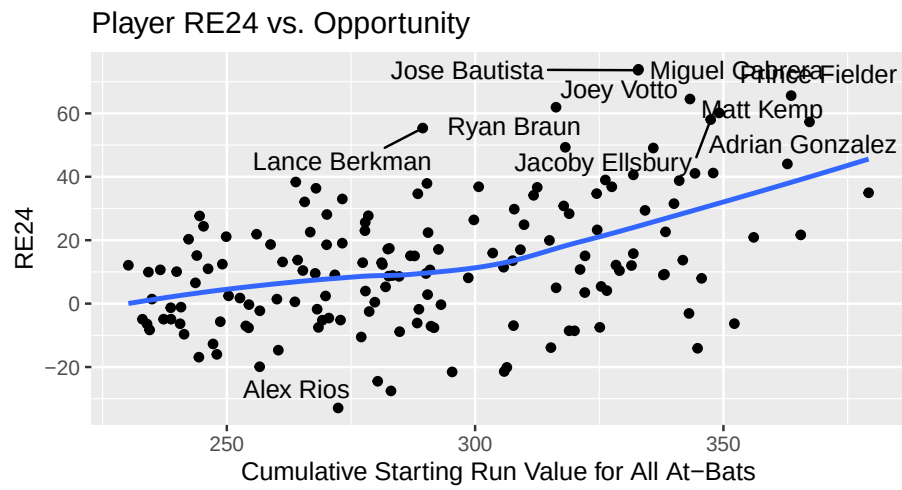


Figure 1: RE24 versus starting runs for all players with at least 502 plate appearances (2011)

Briefly summarize what you would conclude from this chart.

## Value of Hits

**Batting Average** - Traditionally, baseball analysts and fans used batting average to assess players.

$$AVG = \frac{H}{AB}$$

where  $H$  is the number of hits and  $AB$  is the number of at bats.

**Slugging Percentage** - Alternatively, slugging percentage weights the different types of hits by the number of bases the hitter gets.

$$SLG = \frac{X_{1B} + 2 \times X_{2B} + 3 \times X_{3B} + 4 \times HR}{AB}$$

*Discuss strengths and limitations of batting average and slugging percentage in measuring batting performance.*

*How can we use run values to improve these statistics?*

```
# Event Code (variable: EVENT_CD)
# Single: 20
# Double: 21
# Triple: 22
# Home Run: 23

# Create a look-up table to add event names (single, etc) to the data set.
codes <- data.frame(event_cd = 20:23,
```

```

hit_type = factor(c("single", "double", "triple", "home run"),
                  levels = c("single", "double", "triple", "home run"))

codes |>
  kable(caption = "Event codes for each hit type.")

```

Table 1: Event codes for each hit type.

| event_cd | hit_type |
|----------|----------|
| 20       | single   |
| 21       | double   |
| 22       | triple   |
| 23       | home run |

```

# Calculate mean hit values.
mean_hit_values <- retro2011 |>
  filter(event_cd %in% 20:23) |>
  left_join(codes) |>
  group_by(hit_type) |>
  summarize(mean_value = mean(run_value))

mean_hit_values |>
  kable(digits = 2, caption = "Mean run value for each hit type (2011).")

```

Table 2: Mean run value for each hit type (2011).

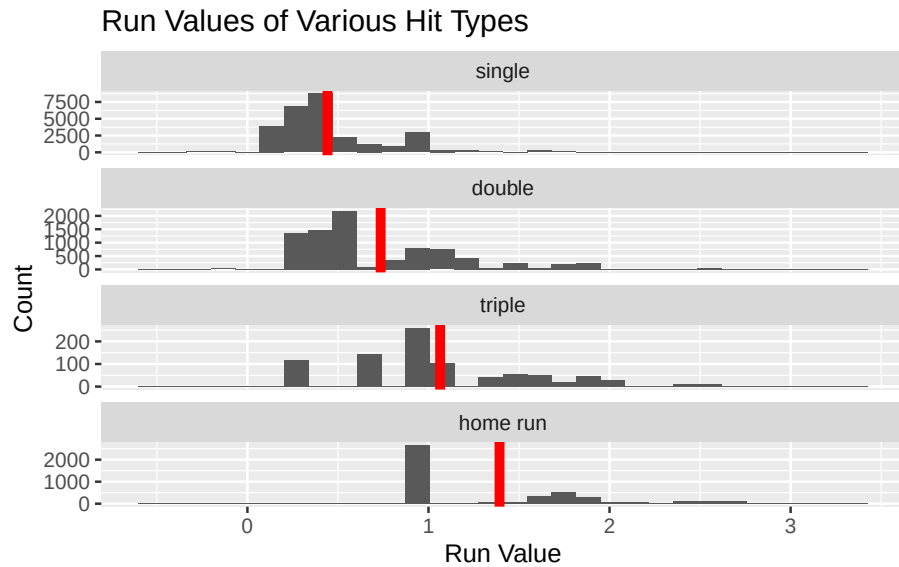
| hit_type | mean_value |
|----------|------------|
| single   | 0.44       |
| double   | 0.74       |
| triple   | 1.06       |
| home run | 1.39       |

```

# Plot run values.
retro2011 |>
  filter(event_cd %in% 20:23) |>
  left_join(codes) |>
  ggplot(aes(x = run_value)) +
  geom_histogram() +
  geom_vline(data = mean_hit_values, aes(xintercept = mean_value), col = "red", size = 2) +
  facet_wrap(~ hit_type, ncol = 1, scales = "free_y") +
  labs(title = "Run Values of Various Hit Types",

```

```
x = "Run Value",
y = "Count")
```



*Based on these results, what would you conclude about AVG and SLG?*

Let's see if we can find players whose slugging percentage is *not* a good predictor of RE24.

```
# Add slugging percentage to the RE24 data frame.
re24_2011 <- Batting |>
  filter(yearID == 2011) |>
  group_by(playerID) |>
  summarize(H = sum(H),
            X2B = sum(X2B),
            X3B = sum(X3B),
            HR = sum(HR),
            AB = sum(AB)) |>
  mutate(X1B = H - X2B - X3B - HR,
         SLG = (X1B + 2*X2B + 3*X3B + 4*HR)/AB) |>
  left_join(select(People, playerID, retroID)) |>
  right_join(re24_2011, by = c("retroID" = "bat_id"))
```

(Random aside: As an alternative to a figure title, consider labeling the figure in a caption. Professional publications often prefer a caption that provides a title and description for a well-labeled figure.)

```
re24_2011 |>
  filter(AB > 400) |>
  ggplot(aes(x = SLG, y = re24)) +
  geom_point() +
  labs(x = "Slugging Percentage",
       y = "RE24")
```

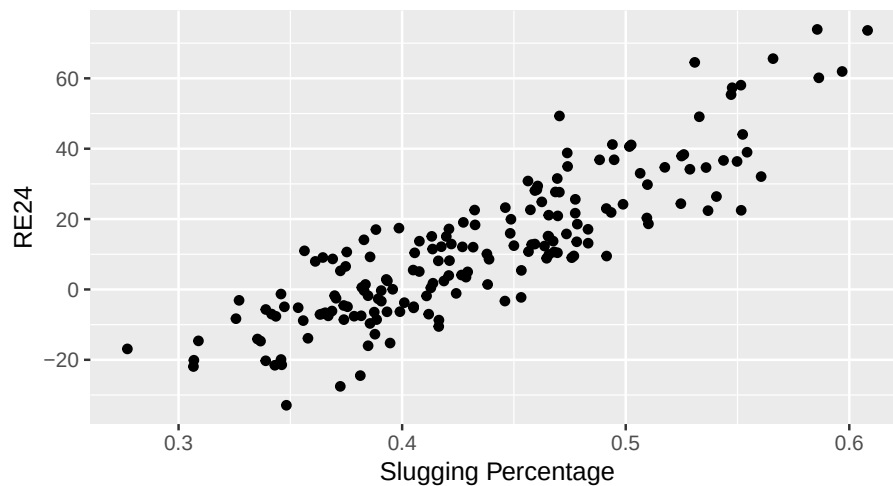


Figure 2: RE24 versus slugging percentage

```
re24_2011 |>
  filter(AB > 400) |>
  arrange(-SLG) |>
  select(name, H, AB, PA, X1B, X2B, X3B, HR, SLG, re24, runs_start) |>
  head(10) |>
  kable()
```

| name          | H   | AB  | PA  | X1B | X2B | X3B | HR | SLG         | re24     | runs_start |
|---------------|-----|-----|-----|-----|-----|-----|----|-------------|----------|------------|
| Jose Bautista | 155 | 513 | 693 | 86  | 24  | 2   | 43 | 0.608187173 | 61702332 | 8156       |
| Ryan Braun    | 187 | 563 | 663 | 110 | 38  | 6   | 33 | 0.596802861 | 94758316 | 2900       |
| Matt Kemp     | 195 | 602 | 704 | 119 | 33  | 4   | 39 | 0.586378760 | 14660349 | 1314       |

| name                 | H   | AB  | PA  | X1B | X2B | X3B | HR | SLG         | re24     | runs__start |
|----------------------|-----|-----|-----|-----|-----|-----|----|-------------|----------|-------------|
| Miguel<br>Cabrera    | 197 | 572 | 701 | 119 | 48  | 0   | 30 | 0.585664373 | 90067332 | 8927        |
| Prince<br>Fielder    | 170 | 569 | 734 | 95  | 36  | 1   | 38 | 0.565905165 | 60064363 | 6424        |
| Adrian<br>Beltre     | 144 | 487 | 543 | 79  | 33  | 0   | 32 | 0.560574932 | 08008265 | 6730        |
| David Ortiz          | 162 | 525 | 627 | 92  | 40  | 1   | 29 | 0.554285738 | 99476326 | 2239        |
| Curtis<br>Granderson | 153 | 583 | 727 | 76  | 26  | 10  | 41 | 0.552315644 | 05812362 | 8833        |
| Pablo<br>Sandoval    | 134 | 426 | 476 | 82  | 26  | 3   | 23 | 0.551643222 | 49106217 | 8158        |
| Jacoby<br>Ellsbury   | 212 | 660 | 748 | 129 | 46  | 5   | 32 | 0.551515258 | 05313347 | 4389        |